VIT
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

School of Computer Science and Engineering

Winter Semester 2023-2024

Continuous Assessment Test – 1

Programme Name &Branch: B.Tech (BCB/BCE/BCI/BCT/BDS/BKT) **Slot : A1+TA1**

Course Name & code: BCSE204L – Design and Analysis of Algorithms

Class Number (s): ALL

Faculty Name (s): ALL

Exam Duration: 90 Mins. **Max. Marks: 50**

General instruction(s): **ANSWER ALL THE QUESTIONS**

<table>
<tr><th>Q.No.</th><th>Question</th></tr>
<tr><td>1.</td><td>(a) Discuss the selection sort algorithm by providing its pseudo code. Discuss the loop invariant in perception with selection sort algorithm. Check the proof of correctness for the same. (4)

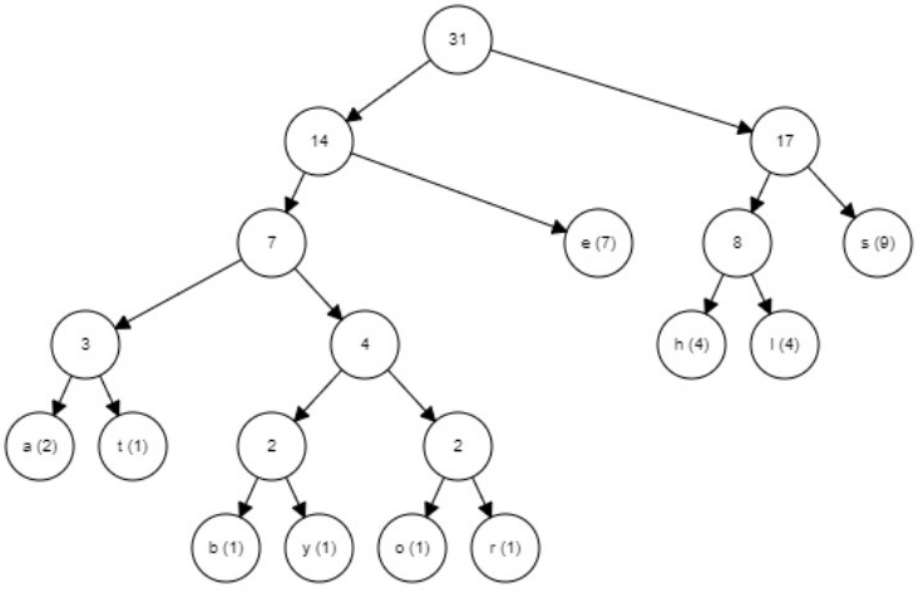
Pseudo code of Selection sort: (1 M)

Loop Invariant: Initialization, Maintenance, and Termination (3M)</td></tr>
<tr><td></td><td>b) Using the master's theorem, solve the recurrence relation
(i) $T(n) = 4T\left(\frac{n}{2}\right) + n^2$
(ii) $T(n) = 7T\left(\frac{n}{2}\right) + 18n^2$ (6M)
(3 M) each
$T(n) = 4T(n/2) + n^2 \Longrightarrow T(n) = \Theta(n^2 \log n)$ (Case 2)
$T(n) = 7T\left(\frac{n}{2}\right) + 18n^2 \Rightarrow T(n) = \Theta\left(n^{log_2 7}\right)$ (Case 1)</td></tr>
<tr><td>2.</td><td>Consider the string "shesellsseashellsbytheseashores". Use minimum number of bits for transmitting the said string. Calculate the number of bits used to encode this using Huffman coding technique. Identify the bits required in both fixed-size and variable length encoding.

Frequency Table (2 M)
Huffman tree (3 M)
Huffman code table (3 M)
Bits for fixed-size coding (1 M)
Bits for variable coding (1 M)

<table>
<tr><th>Character</th><th>Freq</th></tr>
<tr><td>S</td><td>9</td></tr>
<tr><td>E</td><td>7</td></tr>
<tr><td>L</td><td>4</td></tr>
<tr><td>H</td><td>4</td></tr>
<tr><td>A</td><td>2</td></tr>
<tr><td>B</td><td>1</td></tr>
</table>
</td></tr>
</table>

O	1
R	1
T	1
Y	1



Fixed length= 31*4 = 124 bits
Variable length = 88 bits

3. Design and develop an algorithm to multiply 2 integers and analyze their time complexity. Illustrate the technique to multiply the numbers 1334, and 1253.

Karatsuba Algorithm – (2 M)
Time complexity – (1 M)
Final answer using recursion – (7 M)

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X = 1334     Y = 1253    n = 2   a = 13      b = 34      c = 12      d = 53
X = 13   Y = 12      n = 1   a = 1   b = 3   c = 1   d = 2
ac = 1,  bd = 6,     e = 5
X = 34   Y = 53      n = 1   a = 3   b = 4   c = 5   d = 3
ac = 15,     bd = 12,    e = 29
X = 47   Y = 65      n = 1   a = 4   b = 7   c = 6   d = 5
X = 11   Y = 11      n = 1   a = 1   b = 1   c = 1   d = 1
ac = 1,  bd = 1,     e = 2
ac = 24,     bd = 35,    e = 62
ac = 156,    bd = 1802,      e = 1097
1671502
```

4. Provide the optimal parenthesization while multiplying the matrices A1, A2, A3, A4, A5 having dimensions mentioned below.

A1	2 × 5

A2	5 × 10
A3	10 × 5
A4	5 × 6
A5	6 × 10

ANS:
M matrix (6 marks)
S matrix (2 marks)
Optimal parenthesization (2 marks)

M[1][2] = 100
M[2][3] = 250
M[3][4] = 300
M[4][5] = 300
M[1][3] = 200
M[2][4] = 400
M[3][5] = 800
M[1][4] = 260
M[2][5] = 700
M[1][5] = 380
The minimum is 380:
Optimal parenthesization is ((((AB)C)D)E)

5. Define 6-Queens problem. Assuming that the queens are placed column-wise in the 6-Queens problem solved using backtracking. Consider the following intermediate state where the queens are attacking each other. Show the steps that involve backtracking to attain the solution where no queen attacks the other.

		Q			
Q					
			Q		Q
	Q				
				Q	

Each step where backtracking is done must be shown along with the necessary diagrams:
Ans:
Definition: (**2 m**)
Back-tracking: (**8 m**)
Students MUST follow COLUMN-wise working out, not ROW-wise.
Backtrack to 3rd column, then follow working out.
The final answer must be

